Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/04/2015
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Flow Boiling and Condensation Experiment (FBCE)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mudawar@ecn.purdue.edu	**Fax:**	FY 765-494-0539
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-5705
Organization Name:	Purdue University		
PI Address 1:	Mechanical Engineering Building		
PI Address 2:	585 Purdue Mall		
PI Web Page:	https://		
City:	West Lafayette	**State:**	IN
Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2012	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	5
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 5/31/2017 per NSSC information (Ed., 6/17/13) NOTE: Change in grant number and to be extended, per D. Chao/GRC (Ed. 3/6/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AB01G ; NNX12AK14G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future space missions, including the manned mission to Mars. The emphasis in two-phase fluid physics is evident from several recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems, and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer and condensation. Parabolic flight experiments and preparations for ISS experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will lay the foundation of development of advanced high power to weight ratio thermal management systems for NASA's future space vehicles and planetary bases.
Task Progress:	The third year of this project included a number of efforts addressing mostly flow boiling and critical heat flux (CHF) in microgravity. This report summarizes findings from this two-part investigation that were completed during the third year. The report is presented in two parts. Part 1 explores flow boiling of FC-72 in microgravity, which is simulated in a series of parabolic flight maneuvers. Flow boiling experiments are conducted in a rectangular channel fitted with two opposite heating walls. The operating conditions include liquid inlet velocities of 0.1 – 1.9 m/s, liquid mass velocities of 224.2 – 3347.5 kg/m2s, and inlet subcoolings ranging from 2.8 – 8.1°C. The study includes both high-speed video analysis of interfacial features and heat transfer measurements. A dominant wavy vapor layer behavior is encountered for most operating conditions. Boiling is sustained mostly in 'wetting fronts' corresponding to contact regions between the wave troughs and the wall, and abated near the wave peaks. During a flight parabola, the heated wall temperatures decrease slightly as the aircraft enters the hypergravity ascent phase, then increase slightly during the microgravity phase, and decrease once again during the hypergravity descent. These temperature variations point to enhancement in flow boiling heat transfer with increasing gravity, and conversely a reduction with µge. While Part 1 explored flow boiling conditions leading to CHF, Part 2 addresses events just before CHF, during the CHF transient, and immediately following CHF. For both single-sided and double-sided heating, interfacial behavior just before CHF is characterized by dominant wavy vapor layers covering the heated walls, where liquid is able to access the walls only in the wetting fronts. CHF is associated with successive lift-off of wetting fronts from the walls, consistent with the Interfacial Lift-off Model, which has been validated extensively in past studies using single-sided heating in both µge and 1 ge (Earth gravity). It is shown this model predicts µge double-sided flow boiling CHF with excellent accuracy. Additionally, the model points to convergence of CHF values for µge and 1 ge for inlet velocities greater than about 1 m/s. Therefore, by maintaining velocities above this threshold this allows designers of space systems to achieve inertia-dominated performance as well as to adopt prior data and correlations developed from terrestrial studies.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
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